CONTROL CHARTS

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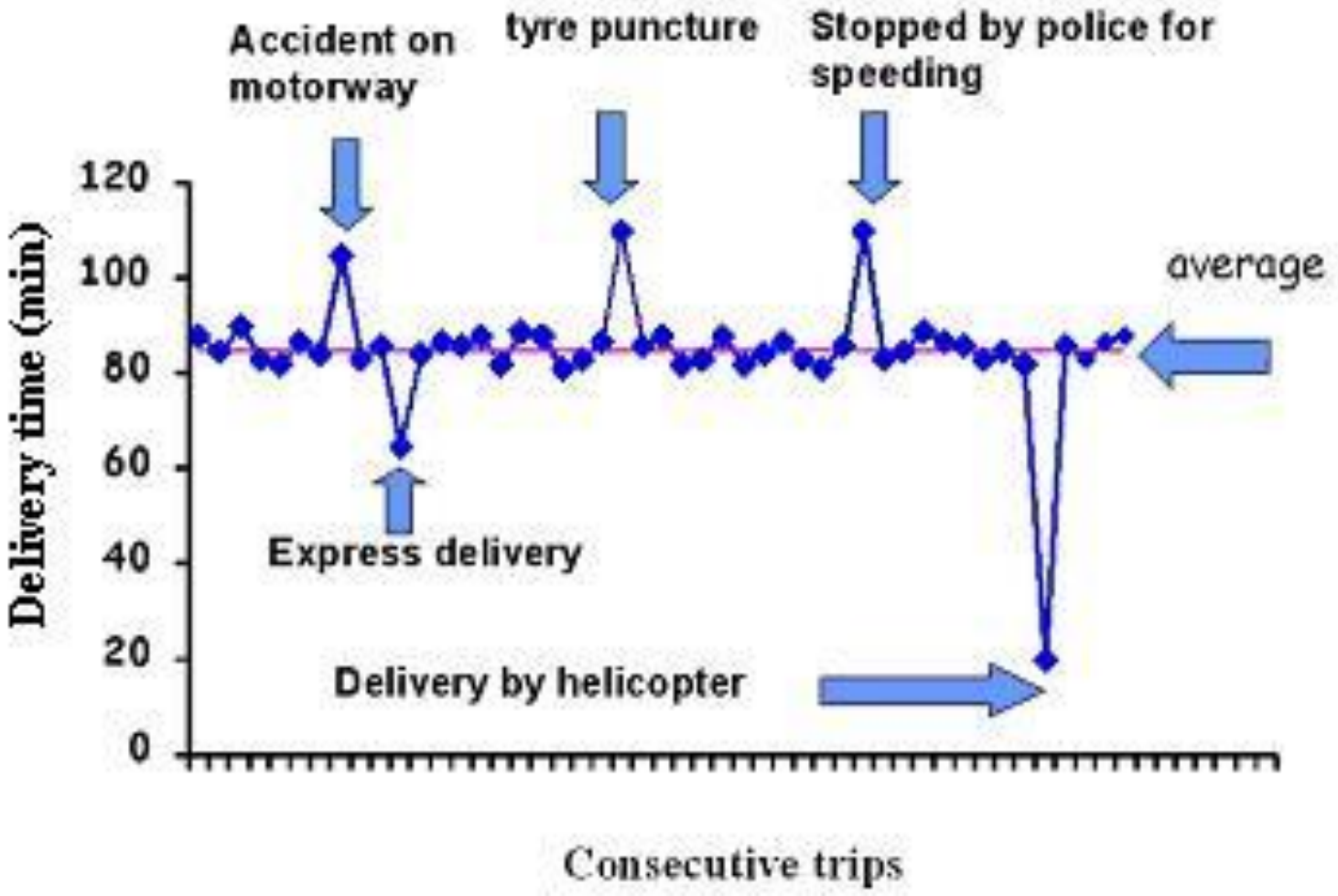
Statistical Process Control

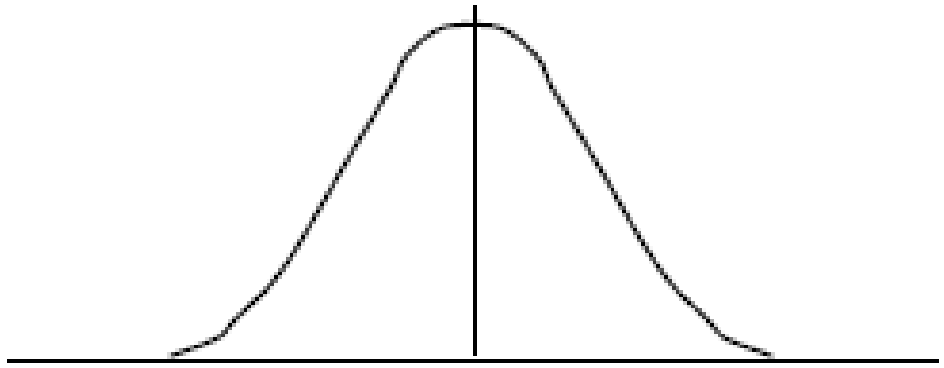
- Statistical process control (SPC) is a concept to solve the problems of quality control.
- The much credit for applying concepts of statistics to problems of quality control goes to Walter Shewhart.
- Shewhart invented the control chart to reduce variability in the performance of telephones at Bell Laboratories
- Edwards Deming recognized the usefulness of these concepts in manufacturing and non-manufacturing sector.

On-line and off-line quality control

- If quality improvement can be attained during the process of production the technique is called on-line quality control.
- Off line quality control refers to the techniques used before the actual process of production starts. Quality is thus built into the product and process as early as at the design stage. This approach eliminates the need for mass inspection that is a wasteful activity.
- In the manufacturing industry, off line quality control is incorporated through *experimental design* methods. Professor Genichi Taguchi of Japan pioneered this method. Also called Taguchi method, it ensures good performance of products or processes by identifying and controlling the factors that contribute to variation.

Assignable and Random Causes





The Central Limit Theorem states that irrespective of the distribution of the individual readings, the sample means are normally distributed with the same mean but narrower standard deviations than that of individual readings.

The distribution of sample means from *any* population tends to a *normal distribution* as the sample size tends to *infinity*. The normal distribution has a variability given by standard deviation (σ). The variability in the sample distribution is given by standard error (σ_x) and related to σ by the relation

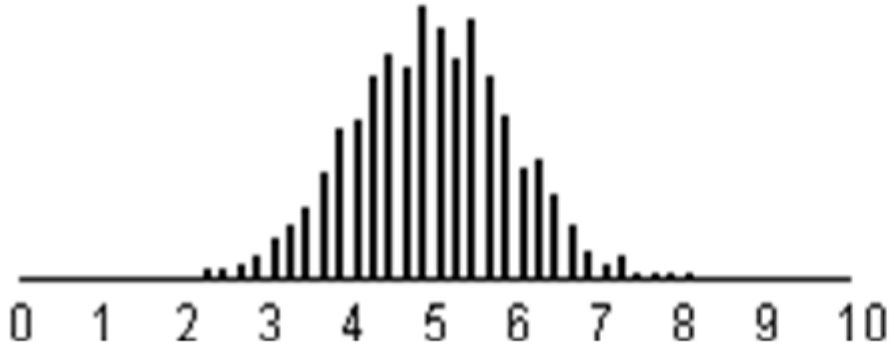
$$\sigma_x = \frac{\sigma}{\sqrt{n}}$$



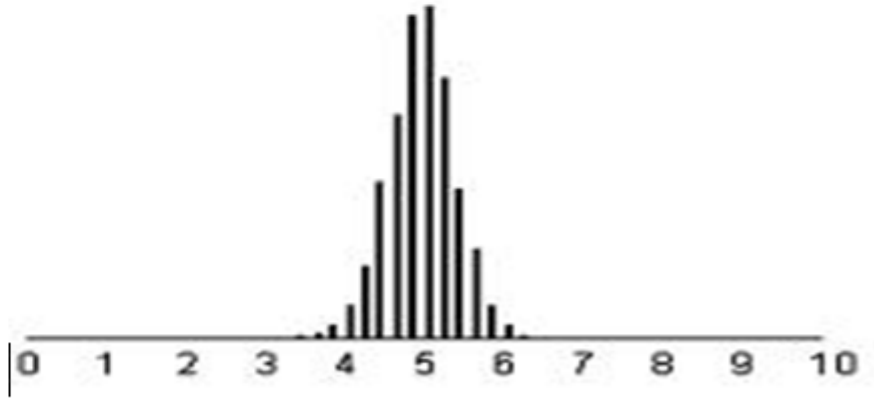
(a) Histogram of Population



(b) $n = 2$



(c) $n = 10$



(d) $n = 50$

Control Chart for Variables

Lower Control Limit= $\mu - 3\frac{\sigma}{\sqrt{n}}$

Upper Control Limit= $\mu + 3\frac{\sigma}{\sqrt{n}}$

The *X*-bar and *R*-charts

1. Record the measurement of k samples of size n. Typically n is 5 and k is at least 20.
2. For each of the k samples, record the mean $\overline{X}_i$
3. Find the range of each sample. R_i = largest – smallest value of each sample, $i = 1 \ldots k$.
4. Calculate the overall average $\overline{\overline{X}}$ and the average range $\overline{R}$ as:

$$\overline{\overline{X}} = \frac{\sum_{i=1}^{k} \overline{X}_i}{k} \qquad \overline{R} = \frac{\sum_{i=1}^{k} R_i}{k}$$

5.

$$LCL = \overline{\overline{X}} - A_2 \overline{R}$$

$$UCL = \overline{\overline{X}} + A_2 \overline{R}$$

n	A_2	D_3	D_4	d_2
2	1.880	0	3.267	1.128
3	1.023	0	2.575	1.693
4	0.729	0	2.282	2.059
5	0.577	0	2.115	2.326
6	0.483	0	2.004	2.534
7	0.419	0.076	1.924	2.704
8	0.373	0.136	1.864	2.847
9	0.337	0.184	1.816	2.970
10	0.308	0.223	1.777	3.078
11	0.285	0.256	1.744	3.173
12	0.266	0.284	1.716	3.258
13	0.249	0.308	1.692	3.336

n	A_2	D_3	D_4	d_2
14	0.235	0.329	1.671	3.407
15	0.223	0.348	1.652	3.472
16	0.212	0.364	1.636	3.532
17	0.203	0.379	1.621	3.588
18	0.194	0.392	1.608	3.640
19	0.187	0.404	1.596	3.689
20	0.180	0.414	1.586	3.735
21	0.173	0.425	1.575	3.778
22	0.167	0.434	1.566	3.819
23	0.162	0.443	1.557	3.858
24	0.157	0.452	1.548	3.895
25	0.153	0.459	1.541	3.931

6. Plot the $\overline{X}$ - chart with *LCL*, *UCL*, centerline and subgroup means ($\overline{X}_i$). This centerline has a magnitude of the overall average $\overline{\overline{X}}$. This line represents the desired average.

7. $$LCL = D_3 \overline{R}$$

$$UCL = D_4 \overline{R}$$

8. Plot the *R*-chart with *LCL*, *UCL*, centerline and subgroup range (R_i). This centerline has a magnitude of average range $\overline{R}$. This line represents a reliable estimate of the range.

9. The process is said to be in statistical control if all subgroup means ($\bar{X}_i$) and ranges (R_i) fall within the respective control limits. Finally, the process standard deviation can be estimated using:

$$\hat{\sigma} = \frac{\bar{R}}{d_2}$$

Suppose we want to construct the X-bar and R-charts for 20 sets of diameter of a piston head. The data is collected from three different measurements. The averages of the three measurements are:

2.0000, 2.0001, 2.0001, 2.0000, 2.0003, 2.0002, 1.9998, 2.0001, 2.0001, 1.9998, 2.0001, 2.0002, 2.0000, 2.0002, 1.9997, 1.9998, 2.0001, 2.0001, 1.9996, 2.0003.

The ranges are obtained as:

0.0004, 0.0005, 0.0007, 0.0007, 0.0001, 0.0004, 0, 0.0001, 0.0006, 0.0006, 0.0003, 0.0007, 0.0003, 0.0004, 0.0007, 0.0010, 0.0006, 0.0001, 0.0004, 0.0008.

$$\overline{\overline{X}} = 2.0000 \text{ and } \overline{R} = 0.0005.$$

From Table corresponding to *n*=3,

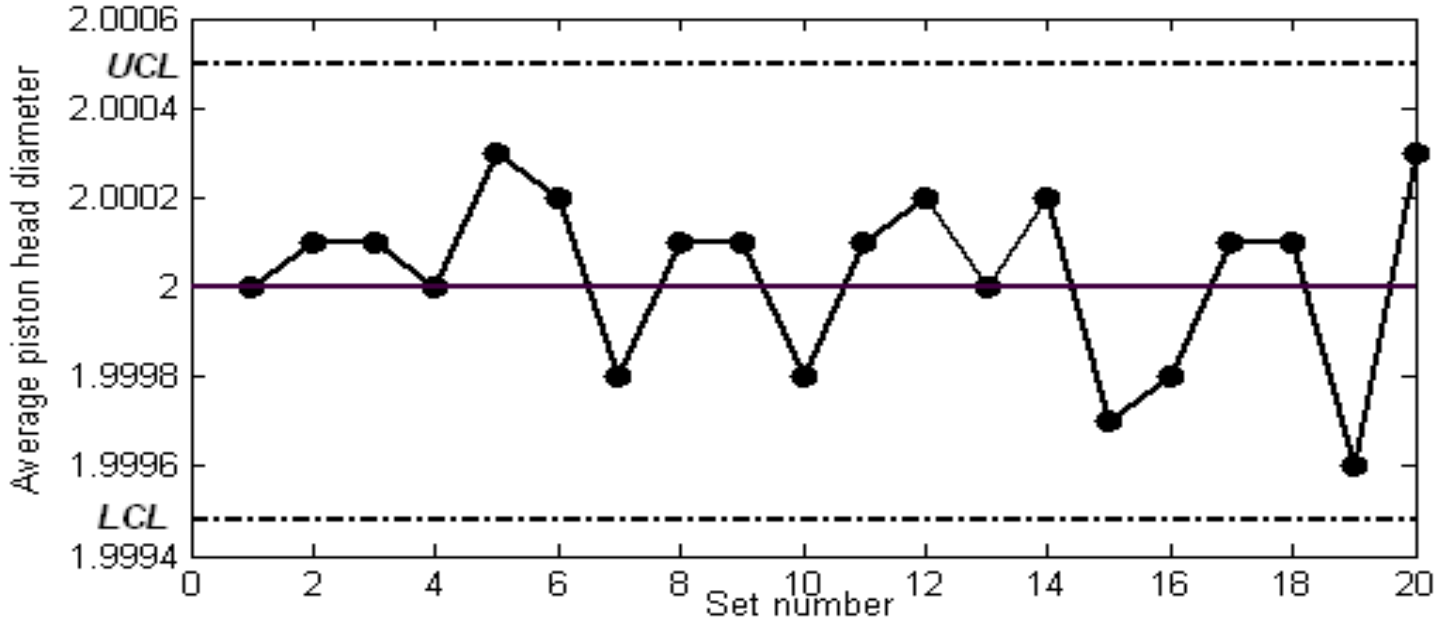
$$A_2 = 1.023,\ D_3 = 0 \text{ and } D_4 = 2.575$$

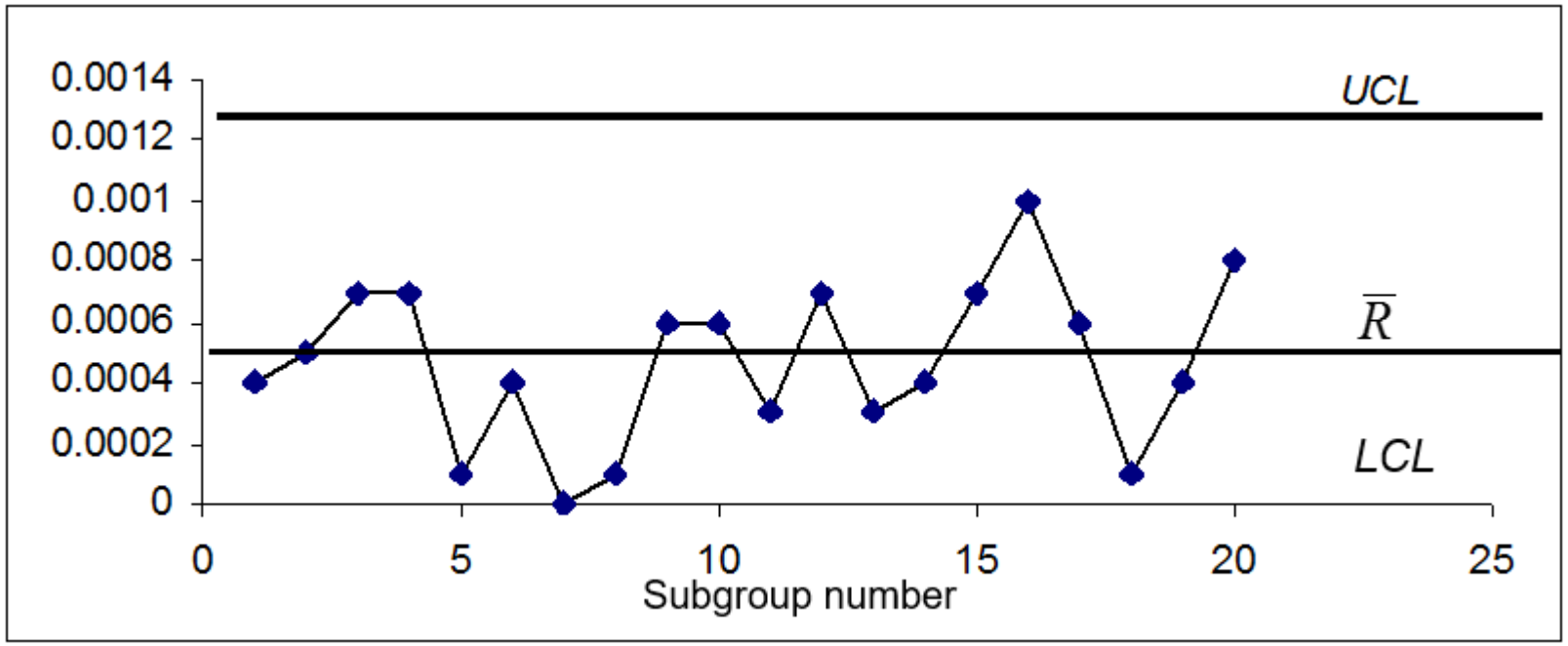
$$LCL = \overline{\overline{X}} - A_2 \overline{R} = 2.0000\text{-}1.023\times 0.0005 = 1.99948$$

$$UCL = \overline{\overline{X}} + A_2 \overline{R} = 2.0000 + 1.023\times 0.0005 = 2.0005.$$

$$LCL = D_3 \overline{R} = 0\times 0.0005 = 0$$

$$UCL = D_4 \overline{R} = 2.575\times 0.0005 = 0.001287.$$


2.0006
UCL
2.0004
2.0002
2
1.9998
1.9996
LCL
1.9994
Average piston head diameter
0
2
4
6
8
10
12
14
16
18
20
Set number


0.0014
0.0012
0.001
0.0008
0.0006
0.0004
0.0002
0
UCL
$\overline{R}$
LCL
0
5
10
15
20
25
Subgroup number

X-bar and s charts

- Instead of range, use standard deviation of sample
- Use when sample size is more say (n>10)
- More precise
- More calculation needed.

CONTROL CHARTS FOR ATTRIBUTES

- The control charts for attributes are essential for those quality characteristics that *cannot* be measured physically, but such characteristics can be classified into one or two classes, usually defectives or non-defectives.
- For example, the presence of holes in an aluminum sheet classifies the item as a defective. When the quality characteristics of an item is measured by attribute, a control chart based on the fraction defective can be used.
- For example, in a process running with a 2% defect rate, a mean of 2 defects per sample requires a sample of at least 100 for recording purposes.

The calculation of control limits for attributes depends on two factors.

(a)Whether *numbers* (for constant size) or *proportions* (for varying sample sizes) are being plotted

(b)Whether *defective* items or *defects* are being considered.

- An item is considered to be defective if it fails to meet a required standard due to presence of defects.
- Thus there are two types of control charts *for defective items* based on whether the sample size is varied (*p*-chart) or constant (*np*-chart).
- Also based on the *presence of defects* charts are constructed for varying sample size (*u*-charts) and constant sample size (*c*-charts).

The *p*-chart

- The p-chart is used for process control of defectives when it is not possible to take a sample of constant sample size because the sizes of samples are varying with time.
- Typical examples include arrival of mails in post-office without the pincode number, scratches in glass sheet, holes in a casting, *etc*.
- In such situations, the control limits are drawn for the average fraction defective $\overline{p}$

$$p = \frac{\text{Number of defectives in each group}}{\text{Number inspected}}$$

$$\bar{p} = \frac{\text{Total number of defectives during period}}{\text{Total number of items inspected during period}}$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\hat{n}}}$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\hat{n}}}$$

If the calculation of lower control limit yields a negative quantity then *LCL* is treated as zero because a negative limit on fraction defective does not makes sense.

S/no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
d	2	3	4	1	0	2	4	1	1	3	0	1	2	1	0	3	7	2	1	2

Sample size 50

The first fraction defective is 2/50 = 0.04.

The other values are:

.06	.08	.02	0	.04	.08	.02	.02	.06	0	.02	.04	.02	0	.06	.14	.04	.02	.04

$$\bar{p} = \frac{\text{Total number of defectives during period}}{\text{Total number of items inspected during period}} = \frac{40}{20 \times 50} = 0.04$$

$$LCL = 0.04 - 3\sqrt{\frac{0.04(1-0.04)}{50}} = -0.440 \sim 0$$

$$UCL = 0.124$$

It is observed that the fraction defective of 17th point is 0.14 and greater than *UCL*. Hence this subgroup having 7 defectives is discarded.

$$\bar{p} = \frac{40 - 7}{19 \times 50} = 0.0347$$

$$LCL = 0.0347 - 3\sqrt{\frac{0.047(1 - 0.0347)}{50}} = -0.430 \sim 0$$

$$UCL = 0.112$$

The np-chart

1. **Center Line (CL):**

$$CL = n\bar{p} = n \cdot \bar{p}$$

2. **Upper Control Limit (UCL):**

$$UCL = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})}$$

3. **Lower Control Limit (LCL):**

$$LCL = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

If LCL < 0, set LCL = 0 (since the number of defectives cannot be negative).

The c-chart

1. **Center Line (CL):**

$$\mathrm{CL} = \bar{c}$$

Where:

- $\bar{c} = \frac{\sum c_i}{n}$
- c_i: Number of defects in the i-th sample
- n: Number of samples

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2. **Upper Control Limit (UCL):**

$$\mathrm{UCL} = \bar{c} + 3\sqrt{\bar{c}}$$

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3. **Lower Control Limit (LCL):**

$$\mathrm{LCL} = \bar{c} - 3\sqrt{\bar{c}}$$

- If $\mathrm{LCL} < 0$, it is set to zero (since the count of defects can't be negative).